

NAHA, Japan

WMO: 479360

Lat: 26.2072N

Lon: 127.6872E

Elev: 164

StdP: 14.61

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	53.6	55.2	36.4	31.8	56.7	38.3	34.2	57.5	27.6	58.1	25.7	59.6	13.7	0	0.484

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.5	90.2	79.4	89.2	79.3	88.4	79.1	81.5	86.7	80.9	86.2	80.4	85.7	12.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.1	157.1	84.6	79.5	154.1	84.4	79.0	151.2	84.2	45.4	86.6	44.8	86.2	44.3	86.0	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.9	24.1	21.3	50.6	92.4	2.1	2.0	49.1	93.9	47.8	95.1	46.6	96.2	45.1	97.7
			43.9	82.2	1.8	1.0	42.6	83.0	41.6	83.5	40.6	84.1	39.3	84.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	74.3	63.0	63.7	66.3	70.9	75.9	81.2	84.7	84.5	82.6	78.4	72.8	66.4
	DBStd	8.58	4.22	4.72	4.65	4.10	3.49	3.29	1.83	1.89	2.35	3.12	3.71	4.26
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	249	90	75	43	5	0	0	0	0	0	0	1	35
	CDD50	8851	403	384	505	628	801	935	1077	1069	979	880	683	507
	CDD65	3626	29	39	83	183	336	485	612	604	529	415	234	77
	CDH74	32764	8	21	87	482	1915	4954	7691	7521	5977	3248	793	67
	CDH80	10620	0	0	1	21	235	1446	3276	3115	1952	536	37	1
Wind	WSAvg	11.7	11.9	11.7	11.7	11.4	10.8	12.1	11.9	11.7	11.9	12.1	11.8	11.9
Precipitation	PrecAvg	81.4	4.4	4.4	5.9	6.5	9.6	10.3	6.9	9.9	8.4	6.5	4.7	4.4
	PrecMax	130.9	10.8	13.2	16.5	20.7	25.1	33.9	20.1	26.5	43.1	27.1	14.4	10.2
	PrecMin	38.0	0.9	0.8	1.1	1.3	0.5	0.8	0.2	0.9	0.9	0.3	0.2	0.0
	PrecStd	19.4	2.2	2.8	3.5	4.1	5.7	6.2	5.8	5.8	7.0	5.1	3.0	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.2	76.4	78.8	82.2	86.0	89.7	91.6	91.7	90.1	87.7	83.1	78.2
		MCWB	68.2	69.5	72.0	75.5	77.7	80.2	79.4	79.8	79.2	77.8	75.1	70.8
	2%	DB	72.9	74.2	76.3	80.3	84.2	88.2	90.3	90.2	88.8	85.9	81.0	75.9
		MCWB	66.3	68.2	69.8	74.2	77.0	79.9	79.3	79.5	78.7	77.1	73.3	68.6
	5%	DB	71.2	72.6	74.6	78.4	82.7	87.2	89.4	89.1	87.7	84.3	79.5	74.1
		MCWB	64.5	67.0	68.6	72.6	76.2	79.7	79.3	79.3	78.3	75.8	72.2	67.0
	10%	DB	69.6	70.9	73.0	76.8	81.0	86.0	88.5	88.1	86.6	82.8	78.1	72.7
		MCWB	63.0	65.4	67.6	71.5	75.2	79.3	79.2	79.1	77.9	75.0	71.3	65.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.6	71.8	73.2	77.0	79.7	81.9	82.3	81.9	81.0	79.5	77.2	73.1
		MCDB	73.9	74.6	76.2	80.5	83.5	86.6	87.1	87.4	86.9	84.6	80.8	76.1
	2%	WB	68.1	70.1	71.9	75.5	78.4	81.1	81.5	81.3	80.2	78.5	75.4	70.8
		MCDB	71.4	73.4	75.1	78.8	82.0	85.5	86.6	86.9	85.9	83.6	79.4	74.4
	5%	WB	66.0	68.2	70.4	74.3	77.4	80.6	80.9	80.7	79.7	77.7	74.0	68.7
		MCDB	70.0	71.5	73.8	77.5	81.1	85.0	86.3	86.4	85.3	82.6	78.3	72.8
	10%	WB	64.1	66.3	68.6	72.8	76.5	80.0	80.3	80.1	79.1	76.8	72.7	67.0
		MCDB	68.6	70.2	72.2	76.1	80.2	84.3	85.9	85.8	84.6	81.6	77.3	71.7
Mean Daily Temperature Range	5% DB	MDBR	7.5	7.8	8.2	8.1	7.6	7.0	7.5	7.5	7.4	7.1	7.2	7.4
		MCDBR	9.5	9.3	9.5	8.9	8.5	7.5	8.1	8.3	8.2	8.1	8.1	8.9
		MCWBR	6.8	6.7	6.5	5.2	4.3	2.9	3.0	3.1	3.4	3.9	4.8	6.1
	5% WB	MCDBR	8.4	8.5	8.6	8.3	7.8	7.2	7.4	7.6	7.5	7.4	7.3	7.8
		MCWBR	7.4	7.7	7.7	5.9	5.0	3.3	3.3	3.3	3.5	3.8	5.1	6.8
Clear-Sky Solar Irradiance	taub	0.457	0.486	0.560	0.568	0.511	0.470	0.470	0.488	0.483	0.482	0.465	0.457	
	taud	2.135	2.074	1.902	1.942	2.167	2.348	2.340	2.278	2.261	2.205	2.210	2.143	
	Ebn at Noon	246	250	238	240	253	262	262	257	256	248	243	240	
	Edh at Noon	43	49	62	61	48	40	40	43	43	43	40	41	
All-Sky Solar Radiation	RadAvg	773	978	1205	1405	1569	1737	2065	1890	1658	1306	956	781	
	RadStd	92	147	113	142	203	157	168	128	126	114	88	77	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	+0.50	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	N/A	N/A	+0.74	+0.95	N/A	N/A	N/A	N/A

Nomenclature: See separate page